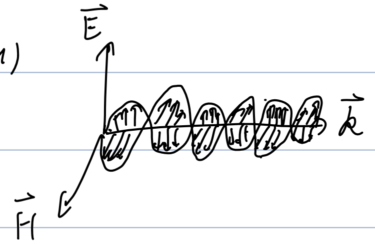
EMC 2024.

Q1.

(a)



Radiation & conduction coupling

(b). From Gauss law, since there's no free charge:

$$\vec{\nabla}\cdot\vec{D} = \frac{\partial}{\partial x}\left(D_m\cos(\omega t-\beta z)\right) = 0$$

$$\vec{\nabla}\cdot\vec{B} = \mu\vec{\nabla}\cdot\vec{H} = \mu\frac{\partial}{\partial y}\left(H_m\cos(\omega t-\beta z)\right) = 0$$

(c) $\Gamma = \frac{Z_m - Z_{air}}{Z_m + Z_{air}}$ $\quad T = \frac{2Z_m}{Z_m + Z_{air}}$

metal

$Z_m = \frac{V(0)}{I(0)} = \frac{V_0^+ + V_0^-}{V_0^+ - V_0^-} Z_0 \Rightarrow V_0^- = \frac{Z_m - Z_0}{Z_L + Z_0} V_0^+$

$\Gamma = \frac{V^-}{V^+} = \frac{Z_m - Z_{air}}{Z_m + Z_{air}}$

$T = 1 + \Gamma = \frac{2Z_m}{Z_m + Z_{air}}$

(d) (i) $Z_m = \sqrt{\frac{j\omega\mu}{\sigma + j\omega\varepsilon}} = \sqrt{\frac{j\cdot 2\pi f\cdot\mu_0}{\sigma + j\cdot 2\pi f\cdot\varepsilon_0}} = \sqrt{2.23\times10^{-4}\angle 90^\circ} \approx 1.49\times10^{-2}\angle 45^\circ\ \Omega$

$Z_{air} = 377\ \Omega$

since $Z_{air} \gg Z_m$, $\Gamma = \frac{Z_m - Z_{air}}{Z_m + Z_{air}} \approx -1$

(ii) $\delta = \sqrt{\frac{2}{2\pi f\cdot\mu\sigma}} = 2.67\times10^{-6}$ m. $\quad \alpha = \frac{1}{\delta} = 3.75\times10^{5}\ m^{-1}$

$20\log e^{\frac{z}{\delta}} = -80 \Rightarrow z = 2.46\times10^{-5}$ m

$\frac{z}{\delta} = \ln 10^4 \qquad \approx 24.6\ \mu m$

Q2.

(a). (i). RF with similar frequency in the ambient space will effect the signal quality.

(ii) Lightning

(iii) Reactive coupling from internal high speed circuit.

Suppress the emission at its source, e.g. reduce high-freq spectral content.

Make the coupling path as inefficient as possible. e.g. shielding

(b) ① use metal shielding. the receptor with a metal enclosure

② isolate RF transmitter with similar frequency band.

Make the receptor less susceptible to the emission

e.g. use of error-correcting codes in a digital receptor.

(c) A: dipole antenna

B: parabolic antenna.

C: patch antenna.

(d) cross-talk. polarization mismatch

Choose one antenna with higher polarization design.

multi-polarized MIMO arrays, circular polarization.

(e) Polarization diversity: by orienting the two dipoles orthogonally

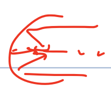
we can see this. or Circular polarization

One for transmition, one for reception

(f). high gain and efficiency. → increase channel capacity

reduce cross talk

directional



For a paraboloids, if we place a source in the focus, the reflected rays will be parallel. The converse is also true.

This leads to its narrow beam, leading to high gain and directicity.

Q3.

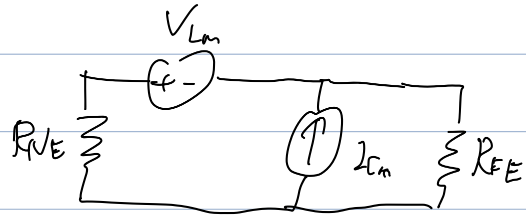
(a).

$V_{Lm} = L\frac{dI}{dt}$

$I_{Cm} = C\cdot\frac{dV}{dt}$

$\frac{dV}{dt} = \frac{10}{40m} \times \frac{100}{150} = \frac{500}{3}$ kV/s

$\frac{dI}{dt} = \frac{250}{150} = \frac{5}{3}$ kA/s



$V_{NE} = \frac{50}{150}V_{Lm} + \frac{5000}{150}\cdot I_{Cm}$

$= \frac{5k}{9}\cdot L + \frac{100}{3}\cdot\frac{500k}{3}\cdot C = 25m$

$V_{FE} = \frac{-100}{150}V_{Cm} + \frac{5000}{150}\cdot I_{Lm}$

$= -\frac{2}{3}\times\frac{5k}{7}\cdot L + \frac{100}{3}\cdot\frac{500k}{3}\cdot C$

$= -\frac{10k}{21}L + \frac{5\times10^4}{9}k\ C = 20m$

$\Rightarrow \begin{cases} L = 4.85\ \mu H \\ C = 4.02\ nF \end{cases}$

(b). Assume $\frac{W}{d} \geq 1$

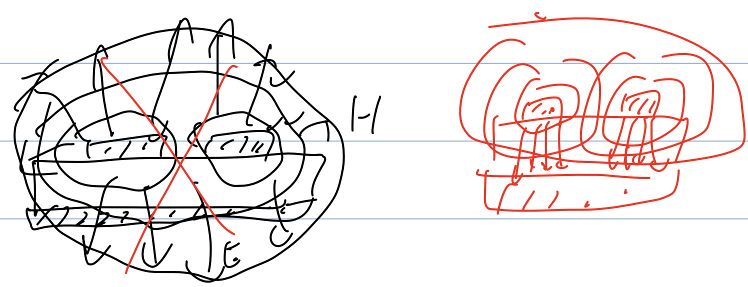
From $Z_0 = \frac{120\pi}{\sqrt{\varepsilon_e}\left[\frac{W}{d} + 1.393 + 0.667\ln\left(\frac{W}{d} + 1.444\right)\right]}$

$\Rightarrow \frac{W}{d} = 1.12$

From $\varepsilon_e = \frac{\varepsilon_r+1}{2} + \frac{\varepsilon_r-1}{2}\cdot\frac{1}{\sqrt{1+12d/W}}$

$\Rightarrow \varepsilon_r = 3.41$

(c)



(d). ① metal box to shield

② ground the plate C

③ place it far from other senstive circuit.

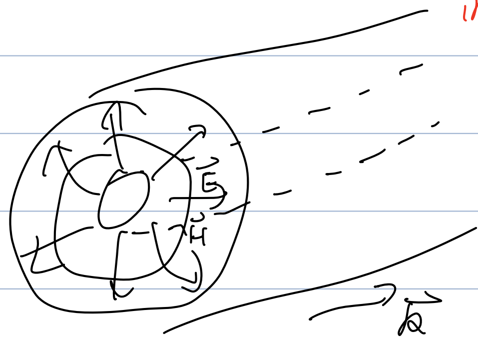
reducing the length of coupling section

increasing gap between TML

increasing the rise and fall time of signal (high-freq)

Q4.

(a).



(b). Highly immune to EMI, since:

① It has sheilding

② will not affect the ambient circuit.

(c). $\delta = \sqrt{\frac{2}{2\pi f \cdot \mu \sigma}} = 6.61\ um.$

$d = 3\delta = 19.8\ um$ is enough to fully immune to most of EMI in the ambient space.

(d) (i) $\Gamma_L = \frac{Z_L - Z_0}{Z_L + Z_0} = \frac{-25+50j}{75+50j} = 0.62\angle 82.9°$

(ii) $\Gamma(z=-0.25\lambda) = \Gamma_0 \cdot \angle(-2\beta d) = 0.62\angle -97.1°$

(iii) $Z_{in} = Z_0 \cdot \frac{Z_L + jZ_0 \tan(\beta l)}{Z_0 + jZ_L \tan(\beta l)} = \frac{Z_0^2}{Z_L} = 20 - 40j = 44.7\angle -63.4°$

(iv). $VSWR = \frac{1+|\Gamma|}{1-|\Gamma|} = 4.26$

(v). When $Z_L = Z_0$, $Z_{in} = Z_0$

Q5.

(a). $\lambda = \frac{c}{f} = 14$ cm

$G_t = 13.5$ dB $= 22.387$

$G_r = 5.3$ dB $= 3.388$

$P_r = P_t \cdot G_t G_r \cdot \left(\frac{\lambda}{2\pi r}\right)^2 = 5.732$ pW $= -112.4$ dBW

-82.4 dBm

(b). (i) ... $\frac{Z_{ant}}{Z_0}$ ...

(ii) $\Gamma = \frac{Z_{ant} - Z_0}{Z_{ant} + Z_0} = 0.59 \angle -104.0^\circ$

(iii): $Z_{in} = Z_0 \cdot \frac{Z_L + jZ_0 \tan\beta l}{Z_0 + jZ_L \tan\beta l} = 100 \cdot \frac{40 - j70 + 100j \cdot \tan 0.6}{100 + j \cdot (40 - j70) \cdot \tan 0.6}$

$= 25.96 - 5.88j\ \Omega = 26.62 \angle -12.76^\circ\ \Omega$

(iv) $\Gamma_{in} = \frac{Z_{in} - Z_0}{Z_{in} + Z_0} = -0.58 - 0.074j = 0.59 \angle -172.8^\circ$.

(v) $VSWR = \frac{1+|\Gamma_{in}|}{1-|\Gamma_{in}|} = 3.88$.

(c).

(i) Other RF signal in the ambient space.

(ii) Lightning

(iii) Reactive coupling from internal high-speed circuit.

Use metal box to shield.

physical hardware: filters

software : coding , encryption

dsp

error correction

$$\vec{\nabla} \times \vec{H} = \vec{J} + \frac{\partial \vec{D}}{\partial t}$$

$$LHS = \begin{vmatrix} \hat{x} & \hat{y} & \hat{z} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ 0 & 0 & D_0 \cos x \sin t \end{vmatrix} = -\hat{y} \frac{\partial}{\partial x}(D_0 \cos x \sin t) = \hat{y} D_0 \sin x \sin t$$

$$RHS = \frac{\partial}{\partial t}(D_0 \sin x \sin t) = D_0 \sin x \cos t$$

no

$$2\pi r \cancel{\ell} E = \frac{\cancel{\ell}\rho}{\varepsilon} \Rightarrow E = \frac{\rho}{2\pi r \varepsilon}$$